Summary in Graph

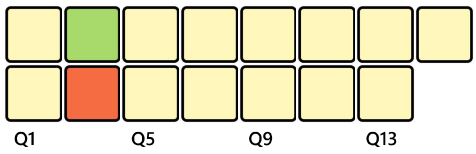
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 10)

Qs. Attempted:	**4** 4 + 0	**Correct Marks:**	**1** 1 + 0
Correct Attempts:	**1** 1 + 0	**Penalty Marks:**	**0.33** 0.33 + 0
Incorrect Attempts:	**3** 3 + 0	**Resultant Marks:**	**0.67** 0.67 + 0

Total Questions:	**15** 10 + 5
Total Marks:	**20** 10 + 10
Exam Duration:	**45 Minutes**
Time Taken:	**45 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (26) out of 255 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

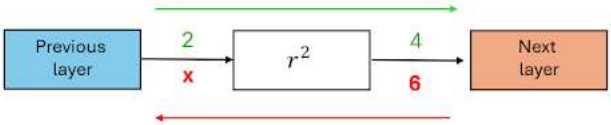
Which of the following statements is true?

A. Adding more hidden layers solves the problem of vanishing gradient for a 2-layer neural network.
B. Adding L2-regularization will help with vanishing gradients
C. Clipping the gradient with a proper threshold solves the problem of exploding gradients
D. Local Gradient for a tanh neuron is upper bounded at 1

Your Answer: A;C;D | Correct Answer: C;D | Incorrect | Time taken: 13min 33sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

The figure below represents a portion of an arbitrary computational graph corresponding to a function approximated by a neural network.



The gradient component ' x ' is

Your Answer: 2.44 | Correct Answer: 24 | Incorrect | Time taken: 10min 02sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Consider a function $y = \text{sigmoid}(z)$ where $z = 2x^3 + 2x^{0.5} - 4$. Assume that these can be drawn into a valid computational graph and the graph is initiated with input = 1. The value of $\left(\frac{\partial y}{\partial z}\right)$ is

Your Answer: .25 | Correct Answer: 0.25 | Correct | Time taken: 05min 14sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Given the regression model $h_\theta(x)$ and the cost function $J(\theta)$, as defined next:
$h_\theta(x) = \theta_3 x^3 + \theta_2 x^2 + \theta_1 x + \theta_0$, where $\theta_3, \theta_2, \theta_1, \theta_0$ are the model parameters
$J(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$, where i iterates over all data, $x^{(i)}$ represents the input of a data point (i), , and $y^{(i)}$ represents the true output of a data point (i)
Which is the partial derivative of the cost function with respect to variable $\theta_3, i.e.,$ $\frac{\partial J(\theta)}{\partial \theta_3}$?

A. $\frac{\partial J(\theta)}{\partial \theta_3} = \frac{2}{n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x^{(i)^3}$
B. $\frac{\partial J(\theta)}{\partial \theta_3} = \frac{1}{2n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^3 x^{(i)^3}$
C. $\frac{\partial J(\theta)}{\partial \theta_3} = \frac{1}{n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x^{(i)}$
D. $\frac{\partial J(\theta)}{\partial \theta_3} = \frac{1}{n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x^{(i)^3}$

Your Answer: D | Correct Answer: A | Incorrect | Time taken: 13min 07sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

An articial neuron has three inputs x_1, x_1, and x_3 . Each input is connected to the neuron with a weight, w1, w2 and w3, respectively. Also, the neuron has a bias b and a ReLU (Rectified Linear Unit) activation function. Assuming that $x_1 = 1, x_2 = 2, x_3 = 0.5, w_1 = 0.5, w_2 = -1, w_3 = -2$, and $b = 1$. which is the neuron output value rounded to three decimals?

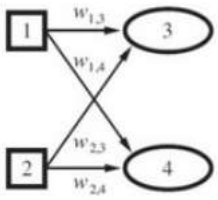
A. 0

B. -1.5
C. 1.5
D. -2.5

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 05min 46sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the Neural Network depicted below, where the dummy ("bias") inputs and weights are not shown!



We assume the following weights: $w_{0,3} = 1, w_{1,3} = -2, w_{2,3} = -1, w_{0,4} = 2, w_{1,4} = -1, w_{2,4} = -2$, The activation function of the neurons 3 and 4 is the **Logistic** function (called the **sigmoid** function in the slides). What is the output of this NN on the input $x = (1, 1)$? First component is the output of neuron 3, second component the output of neuron 4 .
You can find a table values for the **Logistic** function at the last page of the exam.

A. $(0.73, 0.88)$
B. $(0.50, 0.88)$
C. $(0.12, 0.27)$
D. $(0.88, 0.73)$

Assume that sigmoid $(-3) = 0.04$, sigmoid $(0) = 0.5$, sigmoid $(-1) = 0.27$, sigmoid $(-2) = 0.12$, sigmoid(2) $= 0.88$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Assume that we are training a logistic classifier (linear classifier with logistic regression) using the **Logistic** function (called the sigmoid function in the slides) and that the current logistic classifier has the weights $(w_0, w_1, w_2) = (1, 2, -2)$. The next feature point in our training set is given by $x = (1, 2)$. What is the output of this logistic classifier? You can find a table values for the **Logistic** function at the last page of the exam.

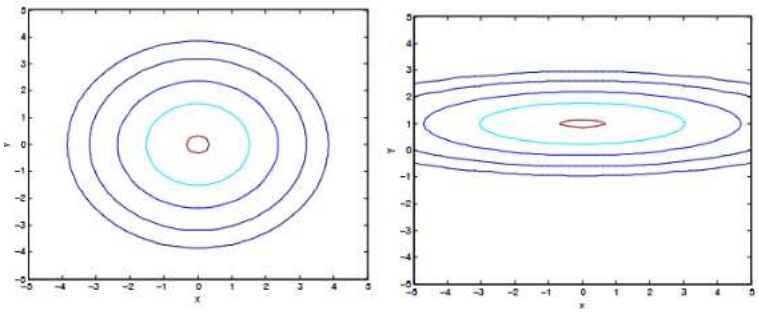
A. 1.00
B. 0.73
C. 0.50
D. 0.27^*

Assume sigmoid $(-1) = 0.27, \text{sigmoid}(0) = 0.5$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

The following diagrams show the iso-probability contours for two different 2D Gaussian distributions. On the left side, the data $\sim N(\mathbf{0}, \mathbf{I})$ where $\mathbf{I}$ is the identity matrix. The right side has the same set of contour levels as left side. What is the mean and covariance matrix for the right side's multivariate Gaussian distribution?



A. $\mu = [0, 0]^T, \quad \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

B. $\mu = [0, 1]^T, \quad \Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 0.25 \end{bmatrix}$

C. $\mu = [0, 1]^T, \quad \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

D. $\mu = [0, 1]^T, \quad \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

With respect to Linear Discriminant Analysis, which of the following is/are true. (Consider a two class case)

A. When both the covariance matrices are spherical and equal, the decision boundary will be the perpendicular bisector of the line joining the means.
B. When both the covariance matrices are spherical and equal, the decision boundary will be perpendicular to the line joining the means.
C. When both the covariance matrices are spherical and equal and the priors $\pi_1 = \pi_2$ then the decision boundary will be perpendicular bisector of the line joining the means.
D. None of the above

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

You are performing LDA on a two-class dataset. You observe that the decision boundary is highly non-linear. What is the most likely reason for this?

A. The features are not normally distributed.
B. The classes have significantly different prior probabilities.
C. The covariance matrices of the classes are vastly different.
D. The dataset has a high dimensionality.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

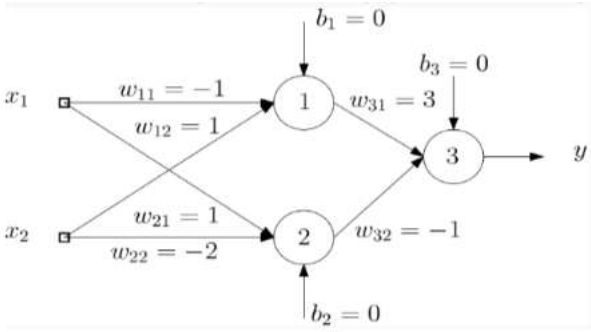
Consider a two-layer feedforward Neural Network. The back-propagation algorithm for this Neural Network consists of several steps:
1: Calculate the δ 's for the output layer.
2: Calculate the output given the input, using forward-propagation.
3: Update the weights of the output layer.
4: Calculate the δ 's of the hidden layer using back-propagation.
5: Update the weights of the hidden layer.
What is the right order of steps?

A. $1, 2, 3, 4, 5$
B. $2, 1, 3, 4, 5$
C. $2, 1, 4, 3, 5$
D. $2, 4, 5, 1, 3$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider a two layered network as in figure given below with input $x = (x_1, x_2)$. Each hidden neuron has the rectified linear unit $h(z) = \max(0, z)$ as 2 points the activation function. Let the input to the network be $x = (1, 2)$ and $d = 1$ be the desired output. The output unit is linear with loss function $J = \frac{1}{2}(y - d)^2$.



Which of the below options are correct ?

A. $\frac{\partial J}{\partial w_{11}}$ is equal to 6
B. $\frac{\partial J}{\partial w_{12}} = \frac{\partial J}{\partial w_{11}} * 2$
C. $\frac{\partial J}{\partial w_{11}} = \frac{\partial J}{\partial w_{12}} * 2$
D. $\frac{\partial J}{\partial w_{11}}$ is equal to 12

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

For a two class classification problem, which among the following are true?

A. In case both the covariance matrices are spherical and equal, the within class variance term has an effect on the LDA derived direction.
B. In case both the covariance matrices are spherical and equal, the within class variance term has no effect on the LDA derived direction.
C. In case both the covariance matrices are spherical but unequal, the within class variance term has an effect on the LDA derived direction.
D. In case both the covariance matrices are spherical but unequal, the within class variance term has no effect on the LDA derived direction.

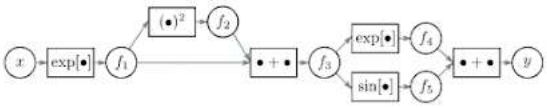
Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a function defined below :

$$y = \exp\left[\exp[x] + \exp[x]^2\right] + \sin\left[\exp[x] + \exp[x]^2\right].$$

We can break above function into a series of intermediate computations so that:



Where f_i is the output form the previous hidden unit.
For example $f_1 = \exp(x)$.
Which of the following options are correct?

A. $\left(\frac{\partial y}{\partial f_2} = \frac{\partial y}{\partial f_3}\right)$
B. $\left(\frac{\partial y}{\partial f_2} = 2 * \frac{\partial y}{\partial f_3}\right)$
C. $2 * \left(\frac{\partial y}{\partial f_2} = \frac{\partial y}{\partial f_3}\right)$
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Given the following details of an articial neuron:

- $\mathbf{x}^\top = [\mathrm{x}_0, \mathrm{x}_1, \mathrm{x}_2]$, a vector collecting all inputs to the neuron, and in which x_0 is the bias.
- $\mathbf{w}^\top = [w_0, w_1, w_2]$, a vector collecting all weights, and in which w_0 corresponds to the bias.
- y, a scalar value representing the true output for an input $\mathbf{x}$
- $f(z) = z$, the activation function
- $\mathrm{o} = \mathrm{f}\left(\mathbf{x}^\top \mathbf{w}\right)$, the output value of the neuron
- Sum of Squared Error (SSE) as the loss function L

If we consider the following data point ($\mathbf{x}^\top = [1, 2, 4], y = 2$), the weights set to $\mathbf{w}^\top = [2, 3, -1]$, and a learning rate of 0.02 , which is the new value of w_2 if we perform one weight update using the gradient descent update rule?

A. 1.16
B. -1.18
C. -1.16
D. -1.32

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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